

IDEAL LOW-PASS DENSITY DESIGN FOR HOMOTOPIC OFF-THE-GRID DECONVOLUTION

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ABSTRACT

A recent conditioning analysis led to a linear program for variable-density continuous frequency sampling. Its solutions empirically equalized a point-spread-function spectrum without dividing the noisy data: the optimized density times the power spectrum was flat on a band. We prove this observation. The continuous problem admits an optimizer whose effective spectrum is a normalized indicator of finitely many intervals; for an even spectrum decreasing away from the origin, it is a centered ideal low-pass. The proof also clarifies non-uniqueness and gives an algorithmic selection rule for the rectangular representative. This shape suggests a coarse-to-fine sequence of nested low-pass variable-projection objectives. A covering-number bound orders objective complexity independently of the optimization method. Equal-acquisition and equal-iteration experiments show a lower separation–SNR transition than direct non-homotopic refinement.

Index Terms— off-the-grid sparse signals, variable-density sampling, homotopy, variable projection, super-resolution

1. INTRODUCTION

Spike deconvolution aims to recover a small number of point sources from a superposition of shifted copies of a known point-spread function (PSF). Unlike discrete sparse recovery, the shifts are real-valued and do not lie on a prescribed grid. This off-the-grid model occurs in radar, array processing, imaging, and channel estimation [1–3]. Random Fourier moments also serve as compact sketches for large-scale mixture learning [4, 5]. In all these examples, Fourier measurements generate a Vandermonde dictionary whose columns become nearly collinear for closely spaced sources, while a low-pass PSF suppresses the frequencies that carry the information needed to distinguish them [6, 7].

A common response is to divide every measurement pointwise by the PSF spectrum. This restores an unweighted spectral-estimation model, but colors and amplifies the noise where the PSF response is weak. The alternative studied here leaves the acquired values untouched and instead designs the frequency-sampling density. The product of that density with the PSF power spectrum is the effective spectrum: its integral measures the average captured energy, and its inverse Fourier transform controls correlations between shifted atoms.

A recent conditioning theorem for variable-density sampling makes this tradeoff quantitative [8]. It bounds the squared singular values of the sensing matrix using the captured spectral energy and

a total-variation penalty scaled by the prescribed minimum separation. A good sampling law must therefore collect energy where the PSF is strong without creating excessive variation in the effective spectrum. Discretizing this criterion gives a convex density-design problem and does not require pointwise equalization.

Numerically, this design repeatedly produced a strikingly simple solution: the effective spectrum was constant on a band and zero outside [8, Sec. 5]. In other words, the sampling law compensated for the PSF statistically and created an ideal low-pass response without dividing any noisy datum by the PSF spectrum. The observation was not covered by the conditioning theorem: it remained unclear whether the rectangular profile was necessarily optimal, whether it was unique, and why a generic linear-program solver sometimes returned a different shape.

The shape question also suggests an optimization strategy. The autocorrelation of a narrow effective band varies slowly; widening the band concentrates it near the origin but introduces more oscillations. This motivates solving a sequence of nested rectangular designs, from a coarse objective to the full-resolution nonconvex objective. Variable projection is well suited to this continuation because it eliminates the source amplitudes and optimizes only over the continuous locations.

The contributions are threefold. First, a complete bounded-variation argument proves the existence of an optimal effective spectrum that is a normalized indicator of finitely many intervals; under the symmetry and monotonicity satisfied by the Gaussian example, a centered interval is optimal. The proof exposes non-uniqueness and yields a deterministic level-set procedure selecting a rectangular optimum. Second, a covering-number bound quantifies how its bandwidth-dependent upper bound grows, without depending on a particular solver. Third, the resulting nested-band continuation is evaluated with variable projection under matched acquisition and total iteration counts, including controls that separate the benefit of the objective path from the initializer.

2. VARIABLE-DENSITY SAMPLING DESIGN

Consider L snapshots sharing the same r locations:

$$x_\ell(t) = \sum_{k=1}^r (\mathbf{A})_{k,\ell} g(t - \tau_k), \quad 1 \leq \ell \leq L, \quad (1)$$

where $\boldsymbol{\tau} = [\tau_1, \dots, \tau_r]^\top \in \mathbb{R}^r$, $\mathbf{A} \in \mathbb{C}^{r \times L}$ collects the amplitudes, and g is the known PSF. The actual minimum separation is $\delta(\boldsymbol{\tau}) := \min_{k \neq k'} |\tau_k - \tau_{k'}| \geq \Delta$. Thus $\delta(\boldsymbol{\tau})$ is the actual minimum separation, whereas Δ is only a prescribed lower bound on it.

Let frequencies $f_1, \dots, f_M$ be drawn i.i.d. from a density ρ supported on $[-B, B]$.

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For each snapshot, we observe the noisy Fourier coefficients

$$(\mathbf{Y})_{j,\ell} := \widehat{x}_\ell(f_j) + (\mathbf{E})_{j,\ell} = \widehat{g}(f_j) \sum_{k=1}^r (\mathbf{A})_{k,\ell} e^{-2\pi i f_j \tau_k} + (\mathbf{E})_{j,\ell}. \quad (2)$$

Define $(\Phi_\tau)_{j,k} = e^{-2\pi i f_j \tau_k}$, $\mathbf{G} = \text{diag}_{1 \leq j \leq M}(\widehat{g}(f_j))$, and stack the entries of (2) in $\mathbf{Y}, \mathbf{E} \in \mathbb{C}^{M \times L}$. The Fourier-domain observation model is therefore $\mathbf{Y} = \mathbf{G}\Phi_\tau \mathbf{A} + \mathbf{E}$. Finally, let $\Gamma_g(\xi) := |\widehat{g}(\xi)|^2$ denote the power spectral density of the PSF.

For the observation model (1)–(2), the result in [8] is stated for one measurement vector. Its high-probability event, however, bounds the singular values of the common sensing matrix $\mathbf{G}\Phi_\tau$. On this event, every $\mathbf{A}$ satisfies

$$\sigma_{\min}^2 \|\mathbf{A}\|_F^2 \leq \|\mathbf{G}\Phi_\tau \mathbf{A}\|_F^2 \leq \sigma_{\max}^2 \|\mathbf{A}\|_F^2,$$

where $\sigma_{\min}$ and $\sigma_{\max}$ are the extreme singular values of $\mathbf{G}\Phi_\tau$. This is the single-vector bound applied to each column and summed over the snapshots. Thus the same high-probability event controls all snapshots: L changes $\mathbf{A}$ and $\mathbf{Y}$, but neither the sensing matrix nor the density and sample-size conditions in [8].

In terms of Γ_g , the theorem assumes $g \in L^1$, $\Gamma_g \in L^1(\rho) \cap \text{TV}(\rho)$, and $\Delta > \|\Gamma_g\|_{\text{TV}(\rho)} / (2\|\Gamma_g\|_{L^1(\rho)})$. For its concentration tolerance $\varepsilon \in (0, 1)$, the stated near-linear sample-size condition then guarantees with high probability, uniformly over all separated τ , that

$$\kappa(\mathbf{G}\Phi_\tau) \leq \left[\frac{1 + \varepsilon \|\Gamma_g\|_{L^1(\rho)} + (2\Delta)^{-1} \|\Gamma_g\|_{\text{TV}(\rho)}}{1 - \varepsilon \|\Gamma_g\|_{L^1(\rho)} - (2\Delta)^{-1} \|\Gamma_g\|_{\text{TV}(\rho)}} \right]^{1/2}. \quad (3)$$

Here

$$\|\Gamma_g\|_{L^1(\rho)} = \int \Gamma_g(\xi) \rho(\xi) d\xi, \quad \|\Gamma_g\|_{\text{TV}(\rho)} = \int_{\mathbb{R}} |d(\Gamma_g \rho)|.$$

The denominator in (3) is the design margin. It makes the tradeoff transparent: placing mass where the PSF is strong increases energy, but abrupt changes of $\Gamma_g \rho$ reduce the uniform separation guarantee.

[8, Sec. 5] discretizes $[-B, B]$ at jB/n and writes $c_j = \Gamma_g(jB/n)$. The proposed design algorithm solves

$$\max_{\substack{\rho \in \mathbb{R}^{2n+1}, \rho \geq 0 \\ (B/n) \sum_{j=-n}^n \rho_j = 1}} u_{L^1}(\rho) - (2\Delta)^{-1} u_{\text{TV}}(\rho),$$

$$u_{L^1}(\rho) = \frac{B}{n} \sum_{j=-n}^n c_j \rho_j,$$

$$u_{\text{TV}}(\rho) = c_{-n} \rho_{-n} + c_n \rho_n + \sum_{j=-n}^{n-1} |c_{j+1} \rho_{j+1} - c_j \rho_j|, \quad (4)$$

The density is then interpolated and used for inverse-CDF frequency sampling. The first two terms of u_{TV} are the jumps to zero at the endpoints. Numerically, the LP produced $c_j \rho_j$ constant on a subset of the grid. The next section proves that this is not a discretization accident.

3. THE OPTIMAL EFFECTIVE SPECTRUM

Set $I = [-B, B]$ and $\lambda = (2\Delta)^{-1}$. For a density ρ , write $h = \Gamma_g \rho$ and extend h by zero outside I . The exact continuous counterpart of (4) is

$$\sup_{\substack{\rho \geq 0 \\ \int_I \rho = 1}} \left\{ \int_I \Gamma_g(\xi) \rho(\xi) d\xi - \lambda \text{TV}(h; \mathbb{R}) \right\}. \quad (5)$$

The next theorem shows that the observed rectangular profile is intrinsic to (5) and identifies when it can be centered.

Theorem 1 (Rectangular effective spectrum) Assume that Γ_g is continuous and positive on I , and $B > \lambda$. Problem (5) admits a

solution. At least one solution satisfies

$$\Gamma_g(\xi) \rho^*(\xi) = \frac{\mathbf{1}_{E^*}(\xi)}{\int_{E^*} \Gamma_g(u)^{-1} du}, \quad (6)$$

where $E^* \subset I$ is a finite union of intervals. If Γ_g is even and strictly decreasing with $|\xi|$, then E^* can be chosen as $[-\Omega^*, \Omega^*]$.

Proof. Put $q = \Gamma_g^{-1}$. Since Γ_g is positive and continuous on the compact interval I , there are constants $0 < q_{\min} \leq q \leq q_{\max}$. In terms of h , the feasible set is

$$\mathcal{A} = \{h \in BV(\mathbb{R}) : h \geq 0, \text{supp } h \subset I, \int_I qh = 1\},$$

and the objective is $J(h) = \int_I h - \lambda |Dh|(\mathbb{R})$.

Existence. The profile $\mathbf{1}_I / \int_I q$ is feasible and has value $(2B - 2\lambda) / \int_I q > 0$; hence the supremum M is positive. Choose a maximizing sequence with $J(h_m) > 0$. The mass constraint gives $\int_I h_m \leq q_{\min}^{-1}$, and therefore

$$|Dh_m|(\mathbb{R}) \leq \lambda^{-1} \int_I h_m \leq (\lambda q_{\min})^{-1}.$$

Thus (h_m) is bounded in BV and has a common compact support. BV compactness yields, after extraction, $h_m \rightarrow h$ in $L^1(\mathbb{R})$ [9, Thm. 3.23]. Nonnegativity, support, and $\int_I qh = 1$ pass to the limit. Lower semicontinuity of total variation then gives $J(h) \geq \limsup_m J(h_m) = M$, so h is a maximizer.

Reduction to a level set. For a finite-perimeter set $E \subset I$, let $Q(E) = \int_E q$. Whenever $Q(E) > 0$, the profile $\mathbf{1}_E / Q(E)$ is feasible; hence

$$|E| - \lambda \text{Per}(E; \mathbb{R}) \leq MQ(E). \quad (7)$$

Apply this inequality to the superlevel sets $E_t = \{h > t\}$. For almost every t , E_t has finite perimeter, and the layer-cake and BV coarea identities [9, Thm. 3.40] give

$$\begin{aligned} \int_0^\infty |E_t| dt &= \int_I h, & \int_0^\infty \text{Per}(E_t; \mathbb{R}) dt &= |Dh|(\mathbb{R}), \\ \int_0^\infty Q(E_t) dt &= \int_I qh = 1. \end{aligned}$$

The gap between the right- and left-hand sides of (7) is nonnegative. Its integral is $M - J(h) = 0$, so it vanishes for almost every nonempty level set. Since $\int_0^\infty Q(E_t) dt = 1$, at least one such set has $Q(E_t) > 0$. Normalizing its indicator gives (6) with value M . A bounded finite-perimeter subset of $\mathbb{R}$ is, up to a null set, a finite union of disjoint intervals.

Centered low-pass representative. Now suppose Γ_g is even and decreasing with $|\xi|$, so q is even and increasing with $|\xi|$. Let E be a level set attaining equality above. The value $|E| - \lambda \text{Per}(E; \mathbb{R}) = MQ(E)$ is positive. The centered interval $E^\#$ with $|E^\#| = |E|$ has $\text{Per}(E^\#; \mathbb{R}) = 2 \leq \text{Per}(E; \mathbb{R})$. Writing $E^\# = [-\Omega, \Omega]$, monotonicity of q also gives

$$\int_{E \setminus E^\#} q \geq q(\Omega) |E \setminus E^\#| = q(\Omega) |E^\# \setminus E| \geq \int_{E^\# \setminus E} q.$$

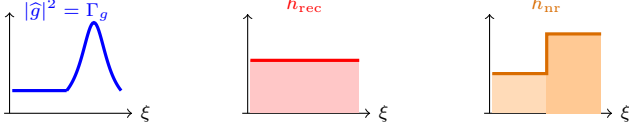
Thus $\int_{E^\#} q \leq \int_E q$. Replacing E by $E^\#$ therefore cannot decrease the objective of the normalized indicator: its numerator does not decrease and its normalizing denominator does not increase. The resulting value is at least M , hence equals M by optimality. This gives an optimal set of the form $[-\Omega^*, \Omega^*]$. $\square$

A concrete non-unique optimum. Consider $I = [-1, 1]$, $\lambda = 1/4$, and a PSF whose power spectral density on I is

$$\Gamma_g(\xi) = q(\xi)^{-1}, \quad q(\xi) = \begin{cases} 1, & -1 \leq \xi \leq 0, \\ 1 - 3\xi + 3\xi^2, & 0 < \xi \leq 1. \end{cases}$$

This is a continuous positive in-band power spectrum. The two effective spectra

$h_{\text{rec}} = \frac{2}{3} \mathbf{1}_I$, $h_{\text{nr}} = \frac{1}{2} \mathbf{1}_{[-1,0)} + \mathbf{1}_{[0,1]}$ both satisfy $\int_I qh = 1$. After zero extension, $J(h_{\text{rec}}) = 4/3 - (1/4)(4/3) = 1$, whereas $J(h_{\text{nr}}) = 3/2 - (1/4)(2) = 1$.



These equal values are globally optimal. Indeed, set $p = 1$ on $[-1, 0]$ and $p(\xi) = 1 - 6\xi^2 + 4\xi^3$ on $[0, 1]$. Then $\|p\|_\infty \leq 1$, and at every jump of either profile, p equals the sign of that jump. Moreover, $\lambda(-p') - 1 + q = 0$. This is the standard subgradient optimality condition for total variation, with multiplier one for the mass constraint, and it certifies both profiles. The rectangle exists, but an LP solver may return another point of the same optimal face.

Selecting the rectangular representative. After solving (4), form $\mathbf{h} = (c_j \rho_j)_j$ and enumerate its distinct positive superlevel sets. Normalize each indicator by the discrete mass constraint and evaluate the LP objective. The discrete layer-cake and coarea identities guarantee that at least one candidate is optimal. Merging consecutive selected cells then returns a minimizer of the form asserted by Thm. 1.

4. NESTED-BAND CONTINUATION

We first define the optimization problem used below. At stage k , let $\mathcal{F}_k$ index the frequencies in $[-\Omega_k, \Omega_k]$, and let $\mathbf{Y}_k$, $\mathbf{G}_k$, and $\Phi_{\tau,k}$ denote the corresponding row restrictions of $\mathbf{Y}$, $\mathbf{G}$, and Φ_τ . For a candidate location vector $\tau \in \mathbb{R}^r$, the amplitudes can be eliminated in closed form:

$$\begin{aligned} \ell_k(\tau) &= \frac{1}{2} \min_{\mathbf{A}} \|\mathbf{Y}_k - \mathbf{G}_k \Phi_{\tau,k} \mathbf{A}\|_F^2 \\ &= \frac{1}{2} \|\mathbf{I} - \mathbf{G}_k \Phi_{\tau,k} (\mathbf{G}_k \Phi_{\tau,k})^\dagger\| \mathbf{Y}_k\|_F^2. \end{aligned} \quad (8)$$

This is the classical variable-projection reduction [10]. For multi-snapshot spike deconvolution, a recent analysis characterizes a bandwidth-dependent basin of convexity and proves local convergence of gradient descent when initialized inside it [11, Thms. 1 and 6].

We can now discuss the geometry behind continuation. When the normalized effective spectrum is uniform on $[-\Omega, \Omega]$, its inverse Fourier transform equals $\text{sinc}(2\Omega t)$, where $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.

Increasing Ω improves resolution but also introduces additional oscillations in the correlations entering (8). The fine loss is therefore more informative while its useful neighborhood around the desired support can be smaller. Solving the smoother problem first and revealing resolution gradually is consequently natural. For the Gaussian design considered below, centered bands satisfy the structural form of Thm. 1; prescribing nested bandwidths produces a controlled path between such objectives.

To quantify complexity independently of the optimizer, let $d\nu_k$ be the probability measure proportional to $\Gamma_g(\xi) \rho_k(\xi) d\xi$, and let

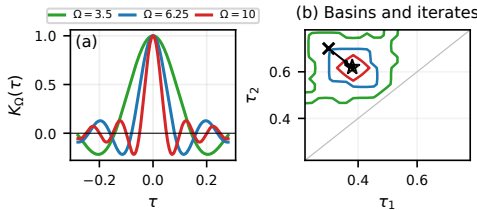


Fig. 1. Coarse-to-fine intuition. (a) Ideal-low-pass autocorrelations for $\Omega = 3.5, 6.25, 10$. (b) Corresponding empirical attraction basins of (8) and nested-band iterates; the star is the true support.

$\beta_k^2 := \int \xi^2 d\nu_k(\xi)$. The corresponding Fourier-feature distance is

$$d_k^2(\tau, \tau') := \int \sum_{s=1}^r \left| e^{-2\pi i \xi \tau_s} - e^{-2\pi i \xi \tau'_s} \right|^2 d\nu_k(\xi). \quad (9)$$

The next proposition isolates the dependence on the effective bandwidth.

Proposition 1 (Effective bandwidth and covering number) *Let $\Theta \subset \mathbb{R}^r$ be a bounded set of ordered location vectors. For every $\varepsilon > 0$,*

$$\mathcal{N}(\Theta, d_k, \varepsilon) \leq \mathcal{N}\left(\Theta, \|\cdot\|_2, \frac{\varepsilon}{2\pi\beta_k}\right).$$

Proof. The scalar mean-value inequality gives $|e^{-2\pi i \xi t} - e^{-2\pi i \xi t'}| \leq 2\pi|\xi||t - t'|$. Applying it componentwise in (9), then integrating against the normalized effective spectrum, yields $d_k(\tau, \tau') \leq 2\pi\beta_k \|\tau - \tau'\|_2$. Thus every Euclidean $\varepsilon/(2\pi\beta_k)$ -net is a d_k -net, which proves the claim. $\square$

For an ideal effective low-pass on $[-\Omega_k, \Omega_k]$, $\beta_k = \Omega_k/\sqrt{3}$. More generally, Prop. 1 shows that $\beta_0 < \dots < \beta_K$ orders the resulting covering-number upper bounds from smaller to larger. It does not assert that every optimization method has nested basins; the VarPro result cited above and Fig. 1 provide the method-specific motivation for warm starts. Together, these two ingredients motivate the coarse-to-fine procedure in Alg. 1.

Alg. 1: Nested-band VarPro continuation

Input: final data $(f_j, (\mathbf{Y})_{j=1}^M)$, bands $\Omega_0 < \dots < \Omega_K = B$.

1. Set $\mathcal{F}_k \leftarrow \{j : |f_j| \leq \Omega_k\}$.
 2. Initialize $\tau^{(0)}$ by standard final-band MUSIC [12].
 3. For $k = 0, \dots, K$, approximately minimize ℓ_k from $\tau^{(k)}$, obtaining $\tau^{(k+1)}$.
- Output:** the final location estimate $\tau^{(K+1)}$.

Alg. 1 reuses one acquisition rather than drawing a new frequency set at every stage. Specifically, draw the M frequencies once from $\rho_B(\xi) \propto \Gamma_g(\xi)^{-1} \mathbf{1}_{\{|\xi| \leq B\}}$. Stage k only retains the already acquired rows indexed by $\mathcal{F}_k$. Conditional on their number, the retained frequencies are i.i.d. from $\rho_{\Omega_k}(\xi) \propto \Gamma_g(\xi)^{-1} \mathbf{1}_{\{|\xi| \leq \Omega_k\}}$. Thus each stage has the intended density for its bandwidth, the row sets are nested, and no frequency is redrawn or newly acquired.

5. NUMERICAL EXPERIMENTS

We use two complementary protocols. Fig. 2 maps successful localization over separation and SNR and compares the resulting 80% frontiers. Fig. 3 fixes the separation and reports localization error together with the corresponding conditional CRBs.

We simulate $L = 20$ independent snapshots of two spikes on the unit torus, with i.i.d. $\mathcal{CN}(0, 1)$ amplitudes, and a Gaussian PS-F $\Gamma_g(\xi) = \exp(-4\pi^2\sigma^2\xi^2)$ with $\sigma = 0.02$. We use $B = 10$, $M = 128$, and bands $\{3.5, 4.75, 6.25, 8, 10\}$, whose effective bandwidths $\beta_k = \Omega_k/\sqrt{3}$ increase from 2.02 to 5.77. Conditional on the frequencies and amplitudes, noise is $\mathbf{E} = \sigma_n \mathbf{Z}$ with i.i.d. $(\mathbf{Z})_{j,\ell} \sim \mathcal{CN}(0, 1)$ and $\sigma_n^2 = \|\mathbf{G}\Phi_\tau \mathbf{A}\|_F^2 10^{-\text{SNR}/10}/(ML)$. Thus conditional expected noise energy, rather than the norm of a particular realization, sets the SNR. Success means permutation-invariant location RMSE below $0.06/(2B)$. Each cell averages 40 trials.

For $0 < \Omega \leq B$, define

$$\rho_\Omega(\xi) = \frac{\Gamma_g(\xi)^{-1} \mathbf{1}_{\{|\xi| \leq \Omega\}}}{\int_{-\Omega}^{\Omega} \Gamma_g(u)^{-1} du}.$$

Each optimized-density trial first draws $f_1, \dots, f_M$ independently from ρ_B . Stage k only masks this acquired array: $\mathcal{F}_k = \{j : |f_j| \leq \Omega_k\}$. Thus $\mathcal{F}_0 \subset \dots \subset \mathcal{F}_K$ exactly; no frequency is redrawn,

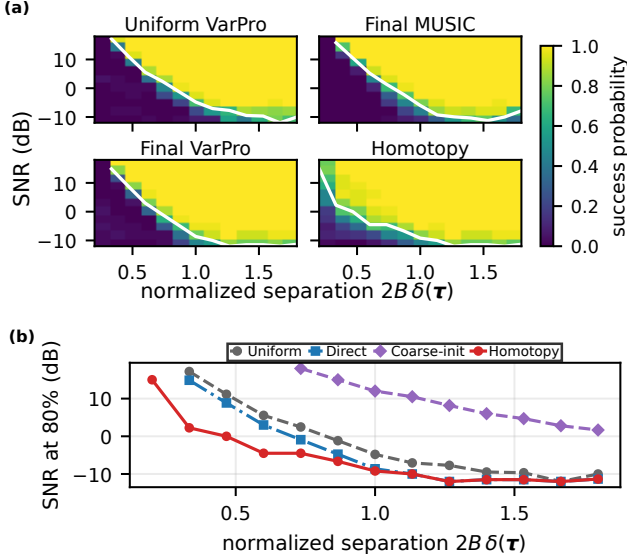


Fig. 2. Empirical success over separation and SNR. (a) Cellwise success probabilities; white curves are the 80% SNR frontiers. (b) The same frontiers overlaid for comparison. Both panels use the same trials and interpolation rule. Abscissa: $2B\delta(\tau)$; lower is better.

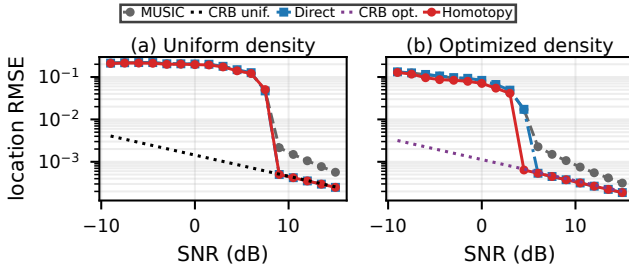


Fig. 3. Location RMSE versus SNR at $2B\delta(\tau) = 0.60$. (a) Uniform; (b) optimized density. Within each panel, the estimators and CRB use the same 80 frequency and amplitude draws. At matched received SNR, the optimized CRB is 0.78 times the uniform CRB. moved, or acquired at a later stage. Conditional on the retained count, these frequencies are i.i.d. with density ρ_{Ω_k} . The corresponding data and noise rows are simply reused. Every method has the same final acquisition count $M = 128$; the uniform baseline draws 128 frequencies uniformly on $[-B, B]$.

All refined methods also receive exactly 60 local VarPro iterations. Direct and homotopic optimized-density methods are paired within every trial: they share the final frequencies, amplitudes, noisy measurements, final-band MUSIC initialization, and final loss. Homotopy allocates (3, 6, 9, 15, 27) iterations across the five nested losses, whereas direct VarPro spends all 60 on ℓ_K . A second non-homotopic control initializes the final loss with coarse-band MUSIC and still spends all 60 iterations at B ; it separates the value of a coarse initializer from that of the intermediate objectives. The uniform baseline uses the same 60-iteration budget. No parameter is tuned per separation–SNR cell.

Fig. 2 gives two views of the same 5,720 Monte Carlo trials. For each method and separation, finite-sample success probabilities may have isolated decreases as SNR increases. We therefore take their cumulative maximum and linearly interpolate the first 80% crossing. Each white curve in (a) is exactly the corresponding colored curve in (b); panel (b) merely overlays the four frontiers.

At $2B\delta(\tau) = 0.60$, homotopy reaches 80% success at -4.50 dB, compared with 3.00 dB for direct optimized-density VarPro and

5.55 dB for uniform sampling. Averaged over separations for which both frontiers cross within the displayed SNR range, the corresponding gains are 2.92 and 5.45 dB. The homotopic curve is never worse than any refined baseline, improves by 18.61 dB on average over the coarse-initialized direct control, and ties direct VarPro in the easy regime beyond $2B\delta(\tau) = 1.13$. The coarse-initialized control shows that transferring a coarse estimate without the intermediate objectives does not reproduce the homotopy gain.

Counting one iteration on one active frequency row as one row-update, homotopy uses only 70.4% of the direct method’s row-updates despite the common 60-iteration budget; its first stage retains 25.2 of the 128 rows on average. Each heatmap cell is the empirical mean of 40 Bernoulli success indicators.

Fig. 3 reports location RMSE at $2B\delta(\tau) = 0.60$ from 80 new paired trials per SNR. Within a panel, all curves share frequencies and amplitudes, and the estimators share each noise realization. Panel (a) uses a uniform density and nested uniform subbands; (b) uses the optimized density and its conditional nested laws.

Under the stated circular-Gaussian noise model, we compute the conditional CRB while treating the complex amplitudes as deterministic nuisance parameters [13]. Eliminating them by a Schur complement gives a location covariance bound. For each trial we take the square root of its mean diagonal entry, and the plotted curve is the root mean square of this quantity over trials. This is a local unbiased benchmark, whereas finite-sample VarPro need not be unbiased. In contrast, the success frontier measures access to the correct basin: a few failures can keep the RMSE high after the 80% crossing in Fig. 2. Using the same success tolerance, the optimized homotopic RMSE first falls below it at 4.5 dB, the other optimized curves at 6 dB, and the uniform curves at 9 dB. At equal received SNR, the optimized CRB is about 22% lower than the uniform one; the larger direct–homotopic gap is therefore algorithmic, not a local-information effect.

6. CONCLUSION

The density-design criterion studied here performs equalization at the level of the sampling law rather than by dividing noisy measurements by the PSF. We proved that its continuous formulation always admits a representative whose effective spectrum is rectangular on finitely many intervals. Under the symmetry and monotonicity satisfied by the Gaussian example, a centered ideal low-pass is optimal. The level-set proof simultaneously explains non-uniqueness: a generic convex solver may return a different point of the optimal face, whereas level-set post-processing deliberately selects a rectangular representative.

This structural result turns a previously empirical observation into a coarse-to-fine design principle. Increasing the ideal-low-pass bandwidth concentrates the autocorrelation, increases our covering-number upper bound, and reveals a finer VarPro landscape. Nested restrictions of one final frequency sample realize the entire path without additional acquisitions. Under equal frequency and total iteration counts, the resulting continuation improves the empirical separation–SNR transition over direct refinement; the coarse-initialization control further shows that the gain comes from the intermediate objectives, not merely from transferring an initializer.

The covering argument is intentionally independent of the optimizer and does not constitute a path-tracking theorem. Establishing conditions under which successive VarPro minimizers remain in the next attraction basin is the main theoretical question left open.

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